

IN625-RAM2 Procurement Scope of Work

N625-RAM2 -AM Alloy Powder Specification – Formulation II Mod 2

1. The IN625-RAM2 -AM powder (formulation II) shall have composition as shown in Table 1.

Name	CAS Number	% Weight
Nickel	7440-02-0	≥50 - ≤75
Iron	7439-89-6	≥10 - ≤20
Chromium	7440-47-3	≥10 - ≤25
Molybdenum	7439-98-7	≥5 - ≤10
Titanium	7440-32-6	≤5
Niobium	7440-03-01	≥1 - ≤5
Silicon	7440-32-6	≤1
Manganese	7439-96-5	≤1
Aluminum Powder (Stabilized)	7429-90-5	≤1
Boron	7440-42-8	≤1
Cobalt	7440-48-4	≤1
Tantalum	7440-25-7	≤1
Copper	7440-50-8	≤1
Carbon	7440-44-0	≤1

Table 1. IN625 RAM2-AM Powder Composition – Formulation II (weight %)

2. The IN625-RAM2 -AM powder particle size distribution (PSD) shall target the ranges listed in Table 2.

Table 2. IN625-RAM2 -AM Powder Size Requirements.

Powder	Mesh or Micron Size	Notes
IN625-RAM2 AM	+325 mesh/1250 mesh (45 um/+10 um)	1. Particles greater than +325 mesh shall be limited to ≤ 1.0 weight % per ASTM B214 2. Particles smaller than 10 (um) shall be limited to ≤ 2.0 cumulative volume % per ASTM B822

3. IN625 RAM2 -AM powder shall be primarily spherical in shape, free of significant satellite particles, and have smooth surfaces.
4. Vendor shall deliver 600 Kgs of IN625 RAM2 -AM within 8 weeks of procurement award.